

RUTU JASWANTH KOTHARI

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EDUCATION

Master of Science in Bioinformatics, Boston University (GPA: 4/4) Dec 2025 (Expected)
Relevant Coursework: R Programming, Applied Machine Learning with Python, Biological Database Analysis, Boston, USA
Translational Bioinformatics, Molecular Biology, Biochemistry, Data Analytics and Interpretation

Bachelor of Engineering in Biotechnology, RV College of Engineering (CGPA: 9.16/10) 2020- 2024
Relevant Coursework: NGS Informatics, Biostatistics, Bioinformatics, Molecular Biology, Biochemistry, Biophysics, Bangalore, India
Engineering Mathematics, Genetic Engineering, Microbial Biotechnology, C++

SKILLS

- Programming Languages:** Python, R, Bash, C++, SQL, MATLAB, Excel, Git
- Data Analysis:** NGS (RNA-seq, CHIP-seq, ATAC-seq, WES), Omics data (Genomics, Transcriptomics, Proteomics), Differential Expression Analysis, Statistical Analysis, Variant calling & annotation, GO Analysis, Machine Learning (Scikit-learn/ PyTorch)
- Pipeline Development and Data Management:** AWS (EC2, S3), Docker, Nextflow, Snakemake, Unix/Linux, HPC, streamlit
- NGS & Bioinformatics Tools:** FastQC, Samtools, Bowtie, GATK, BCFTools, HISAT2, DESeq2, MACS2, Bioconductor, fgsea, BLAST, UCSC Genome Browser, Schrödinger, GROMACS, IUPRED, NCBI, Ensembl, GEO, Human Metabolome, MsigDB
- Data Visualization:** IGV, ggplot2, Plotly, Seaborn, Shiny, VSCode, Streamlit, Tidyverse, Matplotlib

RELEVANT EXPERIENCE

Boston University, Researcher - Lang Lab Present
Boston, MA

- Analyzing epigenomic, transcriptomic, genomic datasets (CUT&RUN, ATAC-seq, RNA-seq, WGS) to uncover gene regulatory mechanisms in stem cells and melanoma using tools such as Bowtie, STAR, DESeq2, and MACS2 for alignment, differential analysis and peak calling.
- Investigating stem cell heterogeneity and differentiation during aging using scRNA-seq (10X Genomics) with pipelines like CellRanger and tools such as Seurat, SCTransform, Splicetools incorporating statistical validation techniques.

Iovance Biotherapeutics, Bioinformatics Intern June 2025 - Sep2025
San Carlos, CA

- Designed scalable, cloud-based pipelines (Docker, AWS, Nextflow) for PARSE split-pool single-cell data, integrating UMI-aware demultiplexing, QC, reproducible workflows for TIL profiling and biomarker discovery.
- Analyzed tumor-derived T cells using scRNA-seq and TCR-seq (VDJ) data with Seurat, scRepertoire, and ScATOMIC, performing clonotype profiling, diversity assessment and doublet detection to support therapeutic development.
- Developed and implemented a logistic regression machine learning model using the Tidymodels framework in R to predict tumor-reactive T cells, enhancing downstream biomarker discovery and immunotherapy profiling.

L&T Technology Services (LTS), Bioinformatics & AI Researcher Jan 2024 - Aug 2024
Bangalore, India

- Integrated multi-omic data (genomic and proteomic) with radiology images, contributing to an AI-powered cancer detection model that improved diagnostic accuracy by 12% and reduced false positives by 7%.
- Implemented MLOps-driven workflows to automate genomic data preprocessing and model updates, increasing model training efficiency by 10% and accelerating deployment in a clinical setting.

Merck Life Science Pvt Ltd, Student Intern April 2023 - Sep 2023
Bangalore, India

- Designed 8 novel carboxamide antibiotics using Schrödinger, PyMOL & ChemDraw, achieving -8.5 kcal/mol binding affinities against E. coli, with chalcone-enhanced compounds increasing antibacterial activity by 30%.
- Applied molecular docking and bioinformatics tools (Bio-Rad CFX Manager, Primer-BLAST) for qPCR analysis, supporting proteomics-related drug discovery, vaccine development and molecular biology workflows.

PROJECTS

Web-Based Omics Data Portal for Female Reproductive Health and LLM Integration

- Built biological database platform using Python (Flask), SQL, JavaScript, AJAX to support intuitive querying, structured storage and interactive visualization of genes, proteins, GO terms, diseases and pathways relevant to the female reproductive system.
- Structured and integrated omics data from public sources (e.g., Ensembl, UniProt, GO) into a normalized schema optimized for knowledge extraction and fine-tuning of large language models (LLMs), enabling biologically contextual search.

Development of R Shiny visualization Application for high-dimensional data analysis

- Developed an R Shiny app for RNA-Seq data analysis for Huntington Disease, featuring Sample Information Exploration, Counts Matrix Exploration, DESeq2-based Differential Expression and GSEA with fgsea.
- Integrated user-friendly interactive features to enable researchers to explore gene expression patterns and pathway enrichment dynamically, improving accessibility and interpretation of high-dimensional RNA-Seq datasets.

Exploring Protein Disorder in Immunogenic Regions Across Species

- Analyzed 10,000+ amino acid sequences from C. elegans, rats, and humans using IUPRED and Python, achieving 90% accuracy in predicting IDRs across 15 immunogenic protein families, with 30% conserved across species.
- Built a homologous protein dataset using UniProt and Reactome, applying machine learning methods like SVM and CNN for pattern recognition and to discover conserved motifs and achieved MCC of 0.85 and ROC AUC of 0.92, advancing drug target identification.

AWARDS AND ACHIEVEMENTS

- Publication:** Kothari, R. J. et al. (2023). *PAX3 Regulatory Signatures and Gene Targets in Melanoma Cells*. **Genes**, 16(5), 577. [PMID: 40428399](#) | DOI: 10.3390/genes16050577
- Best Outgoing Student Award**, RV College of Engineering (2024), Baldwin Girls High School (2018)
- General Merit:** 2nd in Bachelor of Engineering in Biotechnology, RV College of Engineering (2024)
- Leadership:** Captain of university, high school and Karnataka state Basketball teams